

Puzzles Day One

Name: _____

Talk about these mathematical puzzles with a neighbor or in a small group. If you are struggling, ask those around you.

Watermelons:¹ Yesterday a thousand pounds of watermelons lay in the watermelon patch. They were 99% water, but overnight they lost moisture to evaporation and now they are only 98% water. How much do they weigh now? (Hint: since we don't know how many pounds of water there is to start, it may help to use a variable to represent that quantity, say w . What is the initial value of w ?)

Bridge Crossing:² A group of four people has to cross a bridge. It is dark, and they have to light the path with a flashlight. No more than two people can cross the bridge simultaneously, and the group has only one flashlight. It takes different time for the people in the group to cross the bridge:

- Annie crosses the bridge in 1 minute,
- Bob crosses the bridge in 2 minutes,
- Charlie crosses the bridge in 5 minutes,
- Dorothy crosses the bridge in 10 minutes.

How can the entire group reach the other side of the bridge in 17 minutes? (They cannot meet each other half way, throw the flashlight, etc.)

¹Taken from *Mathematical Puzzles* by Peter Winkler

²Taken from www.math.utah.edu/~cherk/puzzles.html

Mountain Ranges: There are exactly 5 mountain ranges of length 3 (these are shown to the right, 3 refers to the length of the base). How many mountains of length 4 are there? What if valleys are allowed?

